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# Clutch Rebounds and Their Relationship to Winning in the NBA

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# Background

- “Clutch” defined as last 5 minutes of a close game ( $\leq 5$ -point margin)
  - Rebounds are key for maintaining possession and controlling tempo
  - Standard rebounding stats ignore game context
  - Our goal: evaluate the relationship between **clutch rebounding** and winning
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## A Clutch Rebound that Saved the Season

- Set up Ray Allen's game-tying three in 2013 Finals
- One rebound changed the outcome of a championship
- Illustrates the game-winning impact of clutch rebounding



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How strong is the  
relationship between  
Clutch Rebound  
Differential (CRD) and  
win percentage?

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## Data & Setup

- Data source: NBA 2024  
Play-by-Play via hoopR
- Filtering:
  - Period  $\geq 4$
  - Last 5 minutes ( $\leq 300$  sec left)
  - Score margin  $\leq 5$
- Total games analyzed: all 2024 regular and post-season games

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## Defining the Metric(CRD/100)

- Clutch Rebound Differential (CRD) = Team's clutch rebounds – Opponent's clutch rebounds
- CRD/100 possessions used to standardize across team pace
- Calculated per game for each team

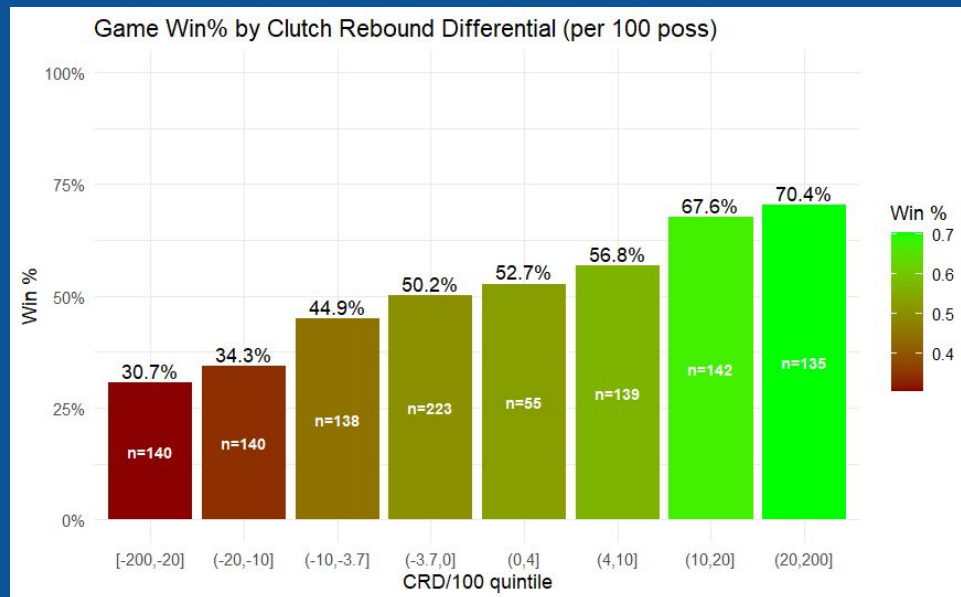
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## Analysis Overview

- Grouped teams into octiles based on average CRD/100
- Calculated game win percentage within each group
- Compared clutch rebounding performance to game win percentage
- Tested both raw CRD and CRD per 100 possessions

## Main Results

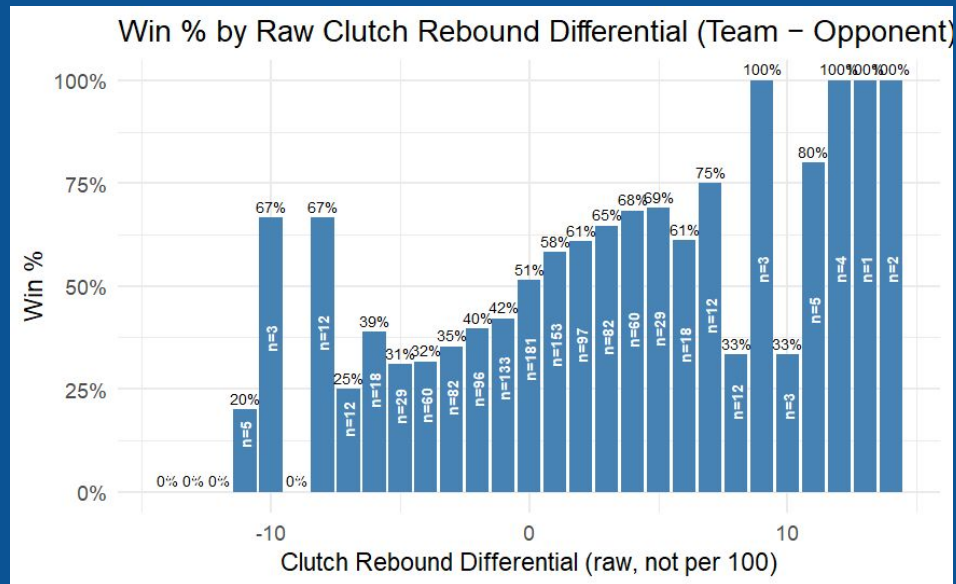
- Each bar represents a range (octile) of CRD/100 values
- As clutch rebound differential increases, win percentage rises
- Teams with low CRD/100 ( $< -20$ ) win ~30% of games
- Teams with high CRD/100 ( $> 10$ ) win ~70% of games
- Shows a clear positive trend between clutch rebounding and winning
- Correlation with team's win percentage in clutch games:  $r \approx 0.39$  (moderate positive)





## Main Results

- Shows win % by raw clutch rebound margin
- Teams with more clutch rebounds tend to win more often
- Positive differential → win rates around 60–70%
- Negative differential → win rates often below 40%



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## Analysis Overview

- Raw CRD shows similar pattern but greater variability
- CRD/100 offers more consistent relationship across teams
- Suggests normalization by possessions provides greater insight into clutch rebounds versus many rebounds near end of game

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## Key Takeaway

- Clutch rebounding correlates positively with winning
  - Scaling by possessions improves compatibility
  - Metric can help evaluate late-game performance trends
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## Future Direction

- Add clutch shooting % and turnovers
  - Analyze player-level clutch stats
  - Compare clutch vs non-clutch rebound patterns
  - Compare regular season CRD/100 and postseason CRD/100 separately to evaluate whether clutch rebounds are more valuable in higher-intensity games
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